European Commission

Framework conditions and funding for AI in Science

Mutual Learning Exercise on National Policies for AI in Science

First thematic report

PSF CHALLENGE

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Report



Research and Innovation

Framework conditions and funding for AI in Science. Mutual Learning Exercise on National Policies for AI in Science

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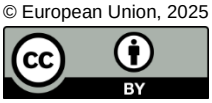
Manuscript completed in April 2025
Final edition

The contents of this publication do not necessarily reflect the position or opinion of the European Commission.

PDF	ISBN 978-92-68-26830-8	doi:10.2777/7211107	KI-01-25-100-EN-N

Luxembourg: Publications Office of the European Union, 2025

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EUROPEAN COMMISSION

Framework conditions and funding for AI in Science

Mutual Learning Exercise on National Policies for AI in Science

First thematic report

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2025

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1. Introduction

The Mutual Learning Exercise (MLE) on National Policies for Artificial Intelligence (AI) in Science provides a forum for Member States and Associated Countries of Horizon Europe to share operational and policy approaches to advance the adoption of AI in science. Building on previous discussions about infrastructure and talent, this report focuses on funding and framework conditions for AI in science.

Drawing from discussions at the MLE workshop in Belgium on 20-21 March 2025, a survey of national policymakers, and desk-based research, this report explores funding, governance, and institutional support models for AI in science from across Europe. It identifies different types of funding frameworks and considers what lessons successful AI in science initiatives offer for the design of funding interventions. It then considers the research governance challenges emerging from AI adoption in science. It concludes by reviewing the diverse policy levers that play a role in enabling AI in science. Across these areas, the report proposes recommendations to support AI adoption in science. Thanks are due to the participants in these evidence-collection activities for sharing insights to inform this document.

1.1. AI as an enabler of scientific discovery

AI is advancing discovery across scientific disciplines.[1] Today, AI applications can be found in all areas of science, technology, engineering, and mathematics (STEM), and in many areas of arts, social science and humanities.[2] These applications span the research lifecycle, with AI tools supporting hypothesis generation, experimental design, and data analysis. Examples of the application of AI in research include:

- **Agricultural science:** By the time that plants have visible symptoms, it may be too late to intervene to prevent a crop from becoming diseased. AI-enabled image analysis offers the possibility of detecting crop disease before plants are visibly affected, enabling earlier interventions that allow farmers to prevent its spread. AI-enabled analysis of spectral data has been used for early disease detection in cassava – a vital food security crop – in Uganda.[3] A smartphone-based image analysis tool offers small holder farmers the possibility of diagnosing plants in-field and responding to the emergence of disease faster than traditional monitoring systems.[4]
- **Climate and Earth sciences:** By processing massive datasets from satellites, sensors, and simulations, AI is enabling better climate and weather forecasting. For example, systems such as GraphCast can provide increased accuracy in predicting extreme weather events up to 10 days in advance.[5] New approaches to end-to-end learning, developed in collaboration with the European Centre for Medium Range Weather Forecasts, are also supporting faster predictive capabilities with lower data and compute needs, opening the possibility of running

[1] Arranz, D., Bianchini, S., Di Girolamo, V., et al. (2023), Trends in the use of AI in science: A bibliometric analysis. Publications Office of the European Union, https://data.europa.eu/doi/10.2777/418191

[2] Berman, B., Chubb, J. and Williams, K. (2024), The use of artificial intelligence in science, technology, engineering, and medicine, The Royal Society, https://royalsociety.org/-/media/policy/projects/science-in-the-age-of-ai/science-ai-taxonomy-report.pdf; Bianchini, S., Müller, M., and Pelletier, P. (2023), Drivers and Barriers of AI Adoption and Use in Scientific Research, arXiv preprint, https://arxiv.org/abs/2312.09843.

[3] Owomugisha, G., Nuwamanya, E., Quinn, J.A., Biehl, M. and Mwebaze, E. (2020), Early detection of plant diseases using spectral data. In Proceedings of the 3rd International Conference on Applications of Intelligent Systems (APPIS 2020), Association for Computing Machinery, New York, NY, USA, Article 26, 1–6, https://doi.org/10.1145/3378184.3378222.

[4] Owomugisha, G. and Mwebaze, E. (2016), Machine Learning for Plant Disease Incidence and Severity Measurements from Leaf Images, *2016 15th IEEE International Conference on Machine Learning and Applications (ICMLA)*, Anaheim, CA, USA, 2016, pp. 158-163, doi: 10.1109/ICMLA.2016.0034.

[5] DeepMind (2024), A new golden age of discovery, https://deepmind.google/public-policy/ai-for-science/.

sophisticated forecasts from desktop computers.[6] In Earth sciences, AI is helping researchers to model tectonic movements and interpret seismic data.[7]

- **Conservation:** AI can help accelerate the synthesis of conservation science research into actionable insights for policymakers. For example, an AI-enabled evidence synthesis tool developed by the Conservation Evidence project filters millions of papers from the OpenAlex dataset to identify conservation-relevant papers, then uses large language models to evaluate their relevance to different research areas. This system can be interrogated by policymakers through an accessible web interface that allows users to explore what academic evidence exists in relation to specific policy questions. The innovation represents one of the first applications of data mining at this scale for conservation literature.[8]

- **Humanities:** Applications of AI in cultural heritage research are creating new understandings of ancient artefacts. For example, research combining cutting-edge machine learning with traditional Assyriology methods is helping to translate thousands of cuneiform tablets from across global museums. This analysis is finding previously unknown social connections that could reveal new insights about family structures and working groups in one of humanity's earliest civilizations.[9]

- **Materials science:** By generating potential material structures and modelling their properties, AI can help researchers computationally explore the viability and desirability of different material structures.[10] AI systems like GNoME have helped identify promising candidates for stable materials,[11] which could unlock progress in a range of areas such as energy storage, electronics, and sustainable manufacturing.

- **Maths**: Conjectures – theories that are likely to be correct but have not been proven – have a storied history of helping researchers to explore new areas of mathematics. AI can play a role in helping generate conjectures to spark mathematical inquiry,[12] and in guiding human intuition in how to solve them.[13]

- **Medicine**: AI tools are being developed to support clinical decision-making. For example, Predict is an online tool developed that helps patients and clinicians assess how different treatments for early invasive breast cancer might improve post-surgical survival rates.[14] Similar AI-driven clinical decision support tools have been developed for other conditions, such as prostate cancer.[15]

- **Structural biology:** AlphaFold has enabled a step-change in protein structure prediction, giving access to 200 million predicted protein structures. This breakthrough allows scientists to explore the function of proteins, which in turn can help develop targeted treatments for

[6] Allen, A., Markou, S., Tebbutt, W. *et al.* (2025), End-to-end data-driven weather prediction, *Nature* (2025), https://doi.org/10.1038/s41586-025-08897-0.

[7] For example: https://github.com/seisbench/seisbench

[8] ai@cam (2025), Conserving with code: How data is helping to save our planet, https://ai.cam.ac.uk/blog/conserving-with-code-how-data-is-helping-to-save-our-planet

[9] Accelerate Science (2024), How can we…develop machine learning tools to map ancient social networks? Blog by Radzim Sendyka, https://science.ai.cam.ac.uk/2024/12/19/how-can-we-develop-machine-learning-tools-to-map-ancient-social-networks.

[10] Merchant, A., Batzner, S., Schoenholz, S.S. et al. (2023), Scaling deep learning for materials discovery, Nature 624, 80–85 (2023), https://doi.org/10.1038/s41586-023-06735-9.

[11] DeepMind (2023), Millions of new materials discovered with deep learning, https://deepmind.google/discover/blog/millions-of-new-materials-discovered-with-deep-learning/.

[12] Mishra, C., Moulik, S.R. and Sarkar, R. (2023), Mathematical conjecture generation using machine intelligence, https://arxiv.org/abs/2306.07277.

[13] Davies, A., Veličković, P., Buesing, L. *et al.* (2021), Advancing mathematics by guiding human intuition with AI. *Nature* **600**, 70–74 (2021), https://doi.org/10.1038/s41586-021-04086-x.

[14] See: https://breast.predict.cam/.

[15] See: https://prostate.predict.cam.

disease, better understanding of biological processes at the molecular level, and further understanding of the interactions between proteins and different drug compounds.[16]

A combination of advances in AI's technical capabilities and growing interest in this area suggest that the coming years may bring accelerating adoption of AI in science. Large language models and generative AI offer researchers a way to engage with complex AI systems using natural language, alongside advanced capabilities that can be used for research tasks such as literature reviews, research planning,[17] or multimodal data analysis in areas like mathematics or biology.[18] Progress in agentic AI offers the possibility of some scientific tasks becoming more automatable.[19] The result is a potential deeper transformation of science, as AI shifts from research assistance to a tool for augmented intelligence that enhances innovation and invention.[20]

1.2. Barriers to AI adoption in science

Despite the success of AI in science in many areas, adoption patterns across domains are variable. Bibliometric analysis suggests that physics, materials science, chemistry, biosciences, and medical sciences show strong adoption, while other fields lag behind.[21] These uneven patterns of adoption reflect a range of technical, institutional, and cultural influences on the use of AI in research. Key barriers to wider adoption of AI in science include:

- **Compute infrastructure:** the development of large models has been a key component of recent progress in AI; model size is correlated with capability, though it is not clear whether this trend will continue.[22] Regardless, this relationship has led to growing demands for access to computational power.[23] While many Member States and the EU are investing in high performance computing (HPC) facilities, the availability and accessibility of computational power remains a barrier to widespread integration of AI in science.
- **Skills and talent:** A thriving AI for science ecosystem would combine cutting-edge AI development with the considered application of AI across research domains and the development of open software tools that enable wider adoption of successful methods. This creates a variety of skill needs, from technical AI capabilities to skills in interdisciplinary collaboration, AI skills for domain scientists, understandings of data management and curation, and software engineering. As demand for these skills grows across the private sector, public sector, and academia, many institutions are grappling with how to respond to skills shortages.[24]
- **Data availability and quality:** Despite long-running efforts to encourage research data management and sharing, access to data is a persistent challenge across research areas.

[16] Jumper, J., Evans, R., Pritzel, A. et al. (2021), Highly accurate protein structure prediction with AlphaFold. *Nature* 596, 583–589 (2021), https://doi.org/10.1038/s41586-021-03819-2.

[17] Van Noorden R, and Perkel JM. (2023), AI and science: what 1,600 researchers think. Nature. Sep;621(7980):672-675. doi: 10.1038/d41586-023-02980-0. PMID: 37758894.

[18] Stanford University (2024), Artificial Intelligence Index Report 2024, Stanford HAI, https://aiindex.stanford.edu/wp-content/uploads/2024/05/HAI_AI-Index-Report-2024.pdf.

[19] Gridach, M., Nanavati, J., Abidine, K.Z., Mendes, L., & Mack, C. (2025), Agentic AI for Scientific Discovery: A Survey of Progress, Challenges, and Future Directions, https://arxiv.org/abs/2503.08979.

[20] Bianchini, S., Müller, M., & Pelletier, P. (2022), Artificial intelligence in science: An emerging general method of invention. Research Policy, 51(10), 104604, https://doi.org/10.1016/j.respol.2022.104604.

[21] Berman, B., Chubb, J. and Williams, K. (2024), The use of artificial intelligence in science, technology, engineering, and medicine, The Royal Society, https://royalsociety.org/-/media/policy/projects/science-in-the-age-of-ai/science-ai-taxonomy-report.pdf; Bianchini, S., Müller, M., and Pelletier, P. (2023), Drivers and Barriers of AI Adoption and Use in Scientific Research, arXiv preprint, https://arxiv.org/abs/2312.09843.

[22] Sevilla, J., Heim, L., Ho, A., Besiroglu, T., Hobbhahn, M., & Villalobos, P. (2022, July), Compute trends across three eras of machine learning. In *2022 International Joint Conference on Neural Networks (IJCNN)* (pp. 1-8). IEEE, https://arxiv.org/abs/2202.05924.

[23] AI Now (2023), Computational Power and AI, https://ainowinstitute.org/publication/policy/compute-and-ai.

[24] Groznik, V., (2025), Talent for AI in Science. Mutual Learning Exercise on National Policies for AI in Science: Third Thematic Report, Publications Office of the European Union, https://op.europa.eu/en/publication-detail/-/publication/894ed9ed-1b5d-11f0-b1a3-01aa75ed71a1/language-en.

Issues include the availability of high-quality, open datasets; a lack of incentives for data sharing and management; technical issues with data quality or data access; and a range of ethical or governance concerns.[25]

- **Interdisciplinary collaboration:** The ability to exchange ideas and know-how across disciplines is central to the success of AI in science. Traditional funding structures struggle to accommodate projects that span multiple disciplines, and even when funded, scientists may face challenges in engaging in genuine interdisciplinary collaboration. AI in science projects might fall between the gaps of established funding streams or encounter difficulties securing sustained support for infrastructure. Research culture in many areas disincentivises researchers from collaborating across domains, particularly where promotion structures or publishing practices prioritise single-author contributions over experimentation with new tools or methods. A lack of training opportunities presents a practical barrier for domain researchers engaging with AI tools, while knowledge transfer between AI and domain experts may also be hindered by departmental silos within research institutions or difficulties communicating concepts across disciplinary boundaries.[26]
- **Trust and trustworthiness:** AI integration in science raises a collection of questions about research ethics and governance. Its use impinges on long-standing norms that seek to uphold the integrity of research, including accountability for scientific outputs, reliability of results (particularly where AI tools might be vulnerable to hallucinations), and data governance. These concerns can already be seen in public perceptions of AI in science. For example, a recent Eurobarometer survey showed that 38% of respondents trust scientific research and discoveries created with the help of AI, while 25% distrust this type of research.[27]

These barriers to AI adoption in science are expanded on in Table 1, reflecting discussions across the MLE workshops and in the survey of MLE participants.

Category	Enablers	Challenges
Funding	Long-term funding for interdisciplinary collaborations; Funding for core AI methods; Funding in applied AI across domains; Funding that supports software engineers and data managers.	Overall limitations on funding availability; Ability to bring together high-quality interdisciplinary teams to respond to funding calls; Access to funding for small-scale experimentation; Support for talent and capability-building in funding.
Technical infrastructure	HPC resources; FAIR data infrastructure; Software tools and platforms; Research software engineering support.	Uneven access to HPC; Design of HPC facilities that work for AI for science projects; Fragmented data systems or lack of data sharing; Lack of resources for data management;

[25] ai@cam (2024), Access to data case studies: Lessons for the national data library, https://ai.cam.ac.uk/reports/access-to-data-case-studies.
[26] For example: Institute of Medicine (US) Committee on Building Bridges in the Brain, Behavioral, and Clinical Sciences; Pellmar TC, Eisenberg L, editors. Bridging Disciplines in the Brain, Behavioral, and Clinical Sciences. Washington (DC): National Academies Press (US); 2000. 3, Barriers to Interdisciplinary Research and Training, https://www.ncbi.nlm.nih.gov/books/NBK44876/#.
[27] See Eurobarometer (2025), European citizens' knowledge and attitudes towards science and technology, https://europa.eu/eurobarometer/surveys/detail/3227.

Category	Enablers	Challenges
		Lack of resources for software engineering or recognition of its importance; Lack of incentives for open software development and maintenance; Environmental impact of research and technical infrastructures.
Human capital	Domain experts with AI knowledge; AI specialists developing methods suitable for science; Research software engineers; Data engineers.	Overarching skills shortages; Difficulties retaining talent in academia; Lack of interdisciplinary career progression pathways; Lack of career progression for software or data engineers; High demand for skills across industry and academia, and difficulties competing with salaries in big tech; Ability to access machine learning engineering support; Limited training opportunities.
Cultural and organisational	Support for interdisciplinary collaboration; Space to experiment with new approaches; Responsible research and innovation practices; Open innovation practices, for example support for open source softwarep; Communities of practice in AI for science and peer mentoring networks; Recognition and reward of interdisciplinary contributions.	Tolerance of early failures in interdisciplinary collaboration and experimentation with new approaches; Publication pressure limiting innovation; Time requirements for data preparation and the need for better data governance; Departmental silos and barriers to collaboration; Lack of understanding of how to implement research integrity frameworks in practice.

Table 1: Barriers to AI adoption in science

1.3. AI for science in national policy frameworks

Over 60 countries have published national AI strategies or policies.[28] While representing national priorities or policy approaches, these share many common themes, including: investment in R&D and innovation infrastructure, such as HPC; talent, skills, and the impact of AI on labour markets; data access and governance; industrial strategy; and AI governance.[29] Underneath these headline areas, governments have proposed a range of policy frameworks to support AI adoption.

Most national AI strategies give limited explicit attention to the adoption of AI in science, or how to overcome the barriers set out in Table 1. However, this absence does not reflect a lack of interest or ambition for the use of AI for research and innovation. Support for AI in science can be seen implicitly in several ways:

- Investment in AI R&D is a core feature of national AI strategies. While implementation approaches vary – for example, the development of national AI institutes, the creation of research networks, or support for specific sectors or missions – funding is an important enabler of AI development and adoption in science.[30]
- Many strategies refer to priority sectors for AI deployment, with the implication that AI should be an enabler of innovation in those sectors. Common priority domains are agriculture and food, energy, mobility and transport, and life sciences and health, though these vary across countries.[31]
- There is a range of policy interventions that are necessary to support AI adoption in science, even if not labelled as such, for example policies to increase data availability, to build research infrastructure, or to attract talent and support skills development.[32]

Evidence for growing policy interest in the adoption of AI in science can be seen in a range of recent national government initiatives,[33] which are explored across this report. It can also be seen in a constellation of EU-level interventions to promote the uptake of AI in science. For instance, the EU's AI Strategy aims to promote the development and deployment of AI solutions that benefit society and the economy.[34] In pursuit of this goal, the development of a strategy for AI in science and an AI Research Council are two priorities of the new Commission.[35] This forthcoming AI in science policy will seek to accelerate AI adoption and steward its use to preserve scientific integrity

[28]OECD AI Policy Observatory (2024), An overview of national AI strategies and policies, https://goingdigital.oecd.org/data/notes/No14_ToolkitNote_AIStrategies.pdf.

[29] This list is inspired by CIFAR (2018), Building an AI World: Report on National and Regional AI Strategies, https://cifar.ca/cifarnews/2018/12/06/building-an-ai-world-report-on-national-and-regional-ai-strategies/ and Saheb, T. and Saheb, T. (2023), Topical review of artificial intelligence national policies: a mixed method analysis, Technology in Society, Volume 74, 2023, https://doi.org/10.1016/j.techsoc.2023.102316.

[30] Galindo, L., K. Perset and F. Sheeka (2021), "An overview of national AI strategies and policies", *OECD Going Digital Toolkit Notes*, No. 14, OECD Publishing, Paris, https://doi.org/10.1787/c05140d9-en.

[31] Galindo, L., K. Perset and F. Sheeka (2021), "An overview of national AI strategies and policies", *OECD Going Digital Toolkit Notes*, No. 14, OECD Publishing, Paris, https://doi.org/10.1787/c05140d9-en and Saheb, T. and Saheb, T. (2023), Topical review of artificial intelligence national policies: a mixed method analysis, Technology in Society, Volume 74, 2023, https://doi.org/10.1016/j.techsoc.2023.102316.

[32] See Table 1.

[33] For example: the UK's £100M AI for Science initiative funded research hubs in areas like materials discovery and climate science; Germany's Helmholtz Association is investing €23M in foundation models for science; the US Department of Energy's FASST programme is coordinating national laboratories to advance AI tools for scientific discovery.

[34] EU AI Continent Action Plan, available at: https://commission.europa.eu/topics/eu-competitiveness/ai-continent_en.

[35] For an overview of these priorities, see: https://commission.europa.eu/document/download/130e9159-8616-4c29-9f61-04592557cf4c_en?filename=Mission%20letter%20-%20ZAHARIEVA.pdf.

and rigour.[36] Guidelines from the Science Advice Mechanism,[37] upcoming European Council Conclusions on the importance of AI in Science, European Research Area (ERA) Forum,[38] and European Research Council (ERC)[39] have already started a European conversation about the future of AI in science. In the context of growing concern about European competitiveness in deep tech innovation, future policy developments in this area are also expected to connect to the New European Innovation Agenda (NEIA) and European economic security strategy.[40]

In parallel, private sector investment in AI for science is growing rapidly. Tech companies invested approx. $189.2 (approx. €166) billion in AI R&D in 2023.[41] Companies like Google DeepMind, Microsoft Research, and Meta AI are establishing dedicated scientific AI research divisions and pursuing collaborations with academic institutions. In Europe, 2025 saw the launch of the AI Champions initiative,[42] which convenes over 70 organisations with a combined $3+ trillion in public market cap, and which aims to mobilise €200 billion of investment in AI to accelerate adoption in established industries and increase European companies' competitiveness.[43]

These national frameworks invoke a range of different policy agendas, including:

- Focused AI policies or strategies that seek to create the enabling conditions for AI research, innovation, and adoption, and to share the benefits of AI innovation across society.
- Research and innovation (R&I) policies that fund science and research infrastructure.
- Economic and industrial policies that support adoption of new technologies in existing sectors, promote growth of the AI sector, and seek economic competitiveness.
- Education policies that promote AI skills and talent development.

An integrated approach that connects framework conditions, funding mechanisms, and on-the-ground adoption of AI in science is needed, so that policies aiming to support AI-enabled science meaningfully tackle the real-world barriers to adoption set out in Table 1 above.

In that context, this report brings together lessons from major initiatives that are supporting the adoption of AI in science with reflections from policymakers on the design of funding and framework conditions. It starts by exploring how funding frameworks can support AI in science, then considers the translation of governance frameworks into research practices, before developing a framework for AI in science policy.

2. Funding Frameworks and Portfolio Design

2.1. Funding mechanisms for AI in science

Investment in AI research and innovation to enhance national competitiveness is a priority area across most national AI strategies. Broadly-speaking, these investments connect to AI in science through three types of funding: for research and infrastructure; for missions or sectors; and for

[36] Overview of EU AI in science activities, available at: https://research-and-innovation.ec.europa.eu/research-area/industrial-research-and-innovation/artificial-intelligence-ai-science_en.

[37] European Commission's Group of Chief Scientific Advisors (2024), Successful and timely uptake of artificial intelligence in science in the EU: scientific opinion. Brussels: European Commission, https://scientificadvice.eu

[38] https://european-research-area.ec.europa.eu.

[39] ERC (2023), Use and impact of artificial intelligence in the scientific process, https://erc.europa.eu/sites/default/files/2023-12/AI_in_science.pdf.

[40] European Commission: Directorate-General for Research and Innovation (2024), *National policies for AI in science*, Publications Office of the European Union, https://data.europa.eu/doi/10.2777/521143.

[41] Stanford University (2024), Artificial Intelligence Index Report 2024, Stanford HAI, https://aiindex.stanford.edu/wp-content/uploads/2024/05/HAI_AI-Index-Report-2024.pdf.

[42] See: https://aichampions.eu/#letter

[43] For the announcement of this initiative, see: https://ec.europa.eu/commission/presscorner/detail/en/ip_25_467.

public-private partnerships. Box 1 gives a brief overview of examples of funding schemes relevant to AI in science.

For research and infrastructure

- EuroHPC (EU): a €7 billion joint undertaking from 2021-27 to build a European supercomputing infrastructure. This scheme provides essential HPC resources needed for training and running large AI models for scientific applications.
- AI Factories (EU): AI Factories will provide access points to European AI infrastructure for researchers, supporting innovative applications of AI in science and industry.[44] They will be developed around EuroHPC infrastructure, with EuroHPC responsible for developing and operating the HPC facilities. The Commissions AI Continent Action plan aims to set up 13 AI factories and 5 gigafactories with enhanced computing capacity.[45]
- Helmholtz Foundation Model Initiative (Germany): €23 million from the Helmholtz Association for seven research projects developing foundation models for scientific domains, such as climate modelling, medical imaging, and materials research, and for coordination across projects.
- Clusters of Excellence (Germany): €40 million over 7 years from the German Research Foundation for the Tübingen Cluster of Excellence in Machine Learning for Science, which creates an interdisciplinary environment where AI methods are developed in tandem with scientific applications.
- AI for Science Hubs (UK): a UKRI-funded network of specialised research centres, with nine hubs sharing £100 million investment. Each hub has a different domain or technical specialism, with some hubs explicitly focused on applying AI in science (for example, healthcare or chemistry) and others focused on AI methods (for example, generative AI).
- Research Software Engineering Fellowships (UK): £4.5 million from UKRI for up to eight research software engineering fellowships at UK research institutions, with the aim of developing career pathways in research software engineering.
- Natural Science Collections Digitisation Programme (UK): £155 million over ten years for UK museums and research institutions to digitise natural history collections, with the aim of creating rich datasets that can be used to train AI systems for biodiversity, taxonomy, and other natural science applications.
- France 2030 (France): €2.22 billion over five years to develop centres of excellence in AI research and education. For example, the ENACT Cluster at the University of Strasbourg received €67 million to develop a centre focusing on natural language processing, AI for science, and digital health, which combines research activities, education programmes, and industry engagement.
- Netherlands eScience Centre (The Netherlands): this national centre for the development of research software is a joint initiative by SURF and the Dutch Research Council (NWO) that provides computational expertise for researchers to implement AI methods in their work.[46]
- Digital Competence Centres (The Netherlands): three Thematic Digital Competence Centres and 22 Digital Competence Centres support open science projects in Dutch research institutes. Thematic Centres are funded by NWO on behalf of the Ministry of Education, Culture and Science, with each receiving €1.5 million for five years. Their areas of focus are: life sciences and health; natural and engineering sciences; and social sciences and humanities. Complementing these, Local Digital Competence Centres within Dutch research institutes provide general support to researchers within that institute.[47]

[44] Information about AI Factories is available at: https://digital-strategy.ec.europa.eu/en/policies/ai-factories

[45] https://commission.europa.eu/topics/eu-competitiveness/ai-continent_en.

[46] https://www.nwo.nl/en/netherlands-escience-center.

[47] https://tdcc.nl/about-tddc/.

- AIM AT 2030 (Austria): Austria's AI Strategy, coordinated by the Federal Ministry for Education, Science and Research, funds: the Austrian Science Fund's Cluster of Excellence "Bilateral AI" at the University of Linz; the Austrian Academy of Science's Institute for AI in Biomedicine in Vienna; and investments in hardware, for example €36 million for MUSICA (Multi-Site Computing Austria) project, which provides GPU-based high-performance computing.
- Flemish AI Plan (Flanders/Belgium): a €35 million annual investment, with the Flanders AI Research Programme at its core. This programme receives €14 million annually to coordinate AI research across 11 consortium partners, including all five Flemish universities. It brings together 40 research groups with over 300 researchers, and funds 100 PhD students working on interdisciplinary applications in four domains: healthcare, society, industry, and environmental sustainability.
- AI Research Centres (Norway): Norway's Research Council is providing NOK 850 million for four to six AI research centres that will advance interdisciplinary and cross-sectoral AI R&D.[48]
- Aurora (US): The Aurora Exascale Supercomputer is housed at the Argonne Leadership Computing Facility. It was built in partnership with Intel and Hewlett Packard Enterprise, and can perform over a quintillion calculations per second.[49] This is accessible for researchers through the INCITE programme hosted by ALCF.[50]

For missions or sectors

- Destination Earth Initiative (EU): €150 million from the European Commission to create a digital twin of the Earth for climate simulation and prediction.
- AI for Health Mission (UK): £100 million from the UK Government to accelerate AI adoption in the life sciences.
- AI for Green (Austria): part of a €523 million AI-related funding package coordinated via the Austrian Research Promotion Agency (FFG), this scheme aims to further the development of AI and its application to climate challenges.
- Frontiers in Artificial Intelligence for Science, Security and Technology (FASST) (US): the previous US Government Administration proposed to allocate $68 million from the US Department of Energy to support 11 multi-institutional projects that aim to "solidify U.S. leadership in safe and trustworthy artificial intelligence systems for scientific discovery".[51]
- CSIRO AI for Missions Programme (Australia): $25M over four years from Australia's national science agency to "create novel AI research and technologies that will enable Missions to solve Australia's greatest challenges".[52]

For public-private partnerships

- Research Ireland Centres (Ireland): a network of 15 centres that engage an ecosystem of over 800 companies to advance R&I. This is supported by €765.5 million investment from Research Ireland, and employs a cost-share model to leverage private sector funding.
- COMET (Austria): Competence Centres for Excellent Technologies, launched in 2006, bring together research and industry partners to "jointly define a research programme with a medium to long term perspective".[53]

[48] https://www.forskningsradet.no/en/call-for-proposals/2024/mandatory-outline-ai-centres/#:~:text=In%202024%2C%20the%20Research%20Council,and%20start%2Dup%20in%202025.
[49] https://www.anl.gov/aurora.
[50] https://doeleadershipcomputing.org/about/.
[51] https://www.energy.gov/fasst123.
[52] https://www.csiro.au/en/careers/scholarships-student-opportunities/postgraduate-programs-and-scholarships/ai-for-missions-phd-program#:~:text=CSIRO%20has%20invested%20%2425M,to%20solve%20Australia's%20greatest%20challenges.
[53] https://www.ffg.at/en/comet/programme.

- Gefion (Denmark): A partnership between the Novo Nordisk Foundation (DKK 600 million) and the Export and Investment Fund of Denmark (DKK 100 million) is building a Danish supercomputer that will use state-of-the-art NVIDIA hardware and software platforms.[54]
- National AI Research Resource (US): The US National AI Research Resource pilot aims to create "national infrastructure that connects US researchers to computational, data, software, model and training resources they need to participate in AI research", bringing together a range of government departments and private sector providers to deliver research resources and training activities.[55]
- AlphaFold: AlphaFold was a collaboration between Google DeepMind and the European Molecular Biology Laboratory's European Bioinformatics Institute (EMBL-EBI). The resulting AlphaFold database "provides open access to over 200 million protein structure predictions to accelerate scientific research".[56]
- ELIAS Alliance (EU): The ELIAS Alliance "aims to solidify European leadership in AI-driven innovation and scientific advancement" by bringing together AI innovation hubs in eight European cities to bolster knowledge translation between academia and industry in AI for science.

Box 1: Funding schemes supporting AI in science[58]

Many more initiatives could be included in Box 1. Across Europe and beyond, there is a diverse landscape of funding mechanisms that could be leveraged to support AI in science. Each type of funding mechanism offers strengths and limitations:

- Large-scale strategic investments, for example in AI for science hubs, offer a route to building long-term capabilities. However, they concentrate resources in selected domains and require substantial upfront financial commitment.
- Mission-oriented funding initiatives, such as the EU's €150 million Destination Earth or Austria's AI for Green programme, rally research communities around specific challenges like climate science or healthcare, but their impact may remain confined to targeted sectors.
- At the institutional level, smaller-scale initiatives have been effective at building capability from the bottom up. Models like Hungary's AI Ambassador Network, KU Leuven's AI as a Service, and Cambridge's Accelerate Programme (see Box 2) demonstrate how smaller investments can seed new collaborations and create spaces for experimentation. These approaches reduce barriers to AI adoption and respond to researcher needs, though they may struggle with long-term sustainability.
- Public-private partnerships have successfully expanded resources available for AI in science, with AlphaFold standing as the most commonly-cited example, but require significant prior investment in public data assets and domain expertise. Further, there are strategic questions about how to build investment portfolios that effectively leverage public and private funding without creating vulnerabilities in critical scientific infrastructure or privatising publicly funded research assets.

The novelty of this field means there is currently little evidence – in the form of rigorous policy evaluation – on the relative effectiveness of these different funding mechanisms. Which mechanism is most appropriate in which national context will depend on the desired research and

[54] https://novonordiskfonden.dk/en/news/denmark-to-build-one-of-the-worlds-most-powerful-ai-supercomputers-accelerating-solutions-to-societal-challenges/.

[55] https://new.nsf.gov/focus-areas/artificial-intelligence/nairr#nairr-pilot-partners-and-contributors-890.

[56] https://alphafold.ebi.ac.uk.

[57] https://elias-ai.eu/elias-alliance/.

[58] This Box presents examples highlighted in surveys of MLE participants, MLE workshops, and in literature reviews.

policy outcomes in relation to AI in science. Despite this variability, several key factors should be considered when developing funding frameworks for AI in science.

2.2. Design considerations for funding frameworks for AI in science

In an environment where there are growing calls to increase funding for AI in science, policymakers are faced with the challenge of how to allocate limited budgets across competing research priorities and how to maximise the impact of these investments.

Design considerations for AI in science funding frameworks include:

- **Balancing funding approaches:** Funding agencies participating in this MLE describe receiving a growing number of funding applications from researchers proposing to use AI in science. This suggests an increasing 'on the ground' appetite from researchers to use AI. Bottom-up approaches to research funding that are guided by researcher interests allow researchers to drive innovation and are more responsive to emerging areas of opportunity. They may, however, lack strategic direction. In contrast, top-down funding schemes can direct attention to priority areas or shared needs, such as infrastructure, but may miss opportunities for innovation.
- **Supporting interdisciplinary collaboration:** AI in science is highly interdisciplinary, meaning that projects might fall between the gaps of domain-focused funding streams. Project proposals might also be difficult to evaluate effectively, with uncertainty about whether researchers are genuinely collaborating outside of disciplinary silos. The nuances of scheme design and evaluation influence the strength of interdisciplinary collaboration enabled by different funding models.
- **Recognising different types of innovation:** Research in the AI in science ecosystem encompasses a mix of 'exploitation' and 'exploration': some projects focus on exploiting known AI methods, while others are trying new AI approaches and forming new collaborations in a bidirectional exchange of ideas. These projects require different types of support in terms of the nature, focus, and duration of funding provided (for example, in relation to the balance of funding required for people, data, compute, and other research costs).
- **Understanding infrastructure needs:** While HPC investments often dominate conversations about AI research, AI in science requires both this technical infrastructure and the software engineers, AI specialists, and knowledge brokers that enable innovation. Policymakers must balance large-scale infrastructure investments with support for human capital, including often overlooked roles like research software engineers and data managers.
- **Enabling flexibility and sustainability:** There is a tension between traditional funding applications and the speed of change in AI. To negotiate this, funding schemes can focus on desired outcomes and milestones, in place of rigid project plans. Many institutional initiatives struggle with long-term sustainability: seed funding might not translate into sustainable programmes and career pathways for interdisciplinary researchers remain unclear.

2.3. Lessons from institutional approaches to AI in science

Drawing from a mix of public, private, and philanthropic funding sources, research institutions are developing a range of mechanisms for enabling AI adoption. These include:

- Fellowships for early career researchers;
- Training schemes to upskill domain researchers in AI methods;
- Communities of practice in AI for science;
- Support or advisory services that provide technical advice to researchers using AI.

Box 2 sets out five case studies of institutional initiatives to support AI in science. While each differs in scale, approach, institutional, and national context, they show how AI adoption in science requires both researcher-led innovation and institutional support structures that enable experimentation, collaboration, and practical implementation. Lessons from these case studies include the following:

- ***Long-term commitments with implementation flexibility create the conditions for sustained impact.*** Tübingen's seven-year funding cycle provides stability for ambitious research agendas, while offering the flexibility to adjust project directions as opportunities emerge. The Helmholtz Foundation Model initiative similarly benefits from sustained strategic investment but allows individual projects to evolve their approaches as they encounter practical challenges in implementation.
- ***Effective funding portfolios balance investments in technical infrastructure and human capital.*** While computing resources and data platforms are necessary foundations for AI in science, these case studies demonstrate that human capital is equally crucial. KU Leuven's focused investment in a dedicated AI service expert delivers outsized impact by addressing specific implementation barriers. HUN-REN's Ambassador Network illustrates how a distributed model of human support can create significant impact. Cambridge University's AI Clinic similarly provides practical engineering support that bridges the gap between theoretical understanding and successful application.
- ***Create spaces for experimentation and early-stage projects, with progression pathways for successful projects.*** KU Leuven's service model enables researchers to try AI approaches with minimal upfront resource commitment, while Cambridge's Accelerate Science small grants catalyse new collaborations that might later develop into larger initiatives. These low-risk entry points for AI experimentation are crucial for broadening adoption, allowing researchers to explore potential applications before committing substantial resources. Building on these small-scale projects, it is helpful to have progression pathways from small projects to larger initiatives.
- ***Incentivise for two-way disciplinary collaboration over the long-term.*** Tübingen's approach explicitly recognises that machine learning should both fuel discoveries in science and be inspired by scientific challenges. This bidirectional flow of ideas is fostered through structured interactions that bring together researchers from different disciplines. The most productive long-term collaborations emerge when both AI researchers and domain scientists find intellectual value in the partnership. This represents a different type of collaboration to the service model that allows domain researchers to quickly troubleshoot the implementation of AI tools. Both are necessary for the long-term development of the field.

Helmholtz Foundation Model Initiative[59]

The Helmholtz Foundation Model Initiative (HFMI) was launched in 2023, with the aim of addressing complex scientific problems through a combination of advanced AI, domain expertise, and access to data and compute.

The initiative funds seven interdisciplinary projects spanning domains such as climate science, healthcare, protein dynamics, and marine biodiversity. Each project brings together multiple Helmholtz research centres and external partners. These projects are supported with €23 million from Helmholtz's central funding and coordinated through a Synergy Unit.

Several factors have contributed to the initiative's success:

[59] https://www.helmholtz-munich.de/en/newsroom/news-all/artikel/helmholtz-invests-23-million-in-research-on-ai-foundation-models.

- Helmholtz had already built a strong interdisciplinary community through its Information and Data Science Framework, which helped research consortia come together quickly in response to a call for projects.
- Helmholtz benefitted from access to large, shared compute resources necessary for developing foundation models.
- The timing of the HFMI aligned with the availability of funding and growing interest in foundation models, creating momentum for implementation.

The need to acquire sufficient compute time, prepare large and complex datasets, and navigate regulatory challenges related to data sharing and compute resource access have all presented implementation challenges. These experiences highlight the importance of infrastructure readiness and data preparation in successful AI for science initiatives.

The HFMI model offers valuable lessons for AI adoption in science, particularly regarding the importance of building interdisciplinary communities. It demonstrates that successful AI for science initiatives benefit from leveraging networks of expertise and providing flexible support for domain-specific applications.

Accelerate Programme for Scientific Discovery, University of Cambridge[60]

The Accelerate Programme for Scientific Discovery, based at the University of Cambridge, aims to advance scientific research by integrating AI across disciplines. Supported by a donation from Schmidt Sciences, Accelerate Science was established in 2020 and supports three core pillars of activity: research, training, and community-building. Activities include:

- Fellowships: Accelerate has provided five-year career acceleration fellowships for early career researchers, alongside funding to help them establish their research programmes and recruit PhD students. Research areas have included AI for string theory, computational speech analysis to detect psychotic disorders, digital twin models for sleep science, and the application of machine learning to scientific challenges like cell biology and animal behaviour.
- Research grants: Accelerate has funded 33 collaborative research projects that catalyse interdisciplinary collaboration. Its funding scheme has supported research across domains such as cancer research, paleoecology, and legal practice. These grants serve as seed funding to help researchers develop new collaborations and explore novel applications of AI within their disciplines.
- Training: A distinguishing feature of the Accelerate Programme is its emphasis on practical skills development and implementation support. The initiative has built a robust training and networking ecosystem that has provided over 1,200 researchers with access to courses, workshops, and engineering clinics. Training covers topics ranging from large language models and diffusion models to Python programming for scientific applications, with a consistent focus on hands-on learning and practical deployment skills.
- Deployment support: Accelerate's AI Clinic provides practical software engineering support for researchers deploying AI in their work. This practical implementation assistance bridges the gap between theoretical understanding of AI techniques and their successful application in research settings. In 2024 alone, the clinic provided consultancy support for 118 research projects across 35 departments, demonstrating demand for this type of practical assistance.

The Accelerate model demonstrates the importance of complementary interventions that address different barriers to implementing AI in science. It shows how practical implementation support can help researchers to experiment with AI without relying on large-scale resources.

[60] https://science.ai.cam.ac.uk//.

University of Tübingen's Cluster of Excellence in Machine Learning for Science[61]

The University of Tübingen's Cluster of Excellence in Machine Learning for Science was amongst the first major European investments in AI-powered scientific discovery.

The Cluster aims to integrate machine learning in all aspects of scientific discovery and train a new generation of interdisciplinary scientists. It brings together researchers from diverse disciplines including neuroscience, medicine, cognitive science, linguistics, economics, physics, geosciences, philosophy, law, and anthropology. This interdisciplinary approach is central to the Cluster's identity and operations.

The Cluster has created an environment where machine learning and domain science experts collaborate regularly and productively, demonstrated by joint publications across disciplinary boundaries. The initiative has successfully recruited and developed eight group leader positions, all of whom progressed to faculty appointments, indicating strong demand for this expertise. The Cluster has also funded "network projects" that either focus on a machine learning technique applicable across multiple scientific domains or address a scientific question through multiple machine learning approaches.

Funding for the Cluster comes primarily from the German Research Foundation (DFG) through the Excellence Strategy programme, providing approximately €40 million over seven years. This long-term, stable funding allows the Cluster to pursue ambitious research agendas and long-term capability building. Additionally, the University of Tübingen and the state government provide sustainability funding for tenure-track professorships and upfront investment for new faculty positions.

Several factors have contributed to the initiative's success:

- The Cluster invested in communication and engagement mechanisms, including member retreats, "ML meets X" workshops that bring together machine learning experts and domain scientists, and an annual conference on machine learning in science.
- The initiative provides support structures for data management, computing resources, and professional software development.
- The national and state governments and University have provided strong institutional backing.

The Tübingen model demonstrates the impact of investing in early career researchers, the importance of creating venues for interdisciplinary exchange, and the value of institutional support.

KU Leuven AI as a Service[62]

KU Leuven's "AI as a Service" model makes AI expertise accessible for science projects across a major research university. Developed by Leuven AI, the University's AI institute founded in 2020, this service model addresses common barriers to AI adoption in scientific research.

The initiative emerged from Leuven AI's recognition of three recurring obstacles researchers face when attempting to incorporate AI into their work: lack of AI expertise on the domain side, mismatches in objectives between AI experts and domain specialists, and difficulties matching available funding to human resources needs. To address these challenges, Leuven AI created a dedicated job profile called the "AI Service Expert" who is available to all KU Leuven colleagues, allowing them to easily access AI expertise without having to recruit specialised personnel for each project. The aim is to have a catalysing effect; accessible AI

[61] https://uni-tuebingen.de/en/research/core-research/cluster-of-excellence-machine-learning/home/#c634489v.

[62] https://ai.kuleuven.be/news/ai-as-a-service-providing-flexible-access-to-ai-expertise-across-ku-leuven.

support encourages researchers to incorporate AI in their work and allows domain experts to quickly test ideas and spin up projects.

Leuven AI employs the AI service expert and recovers costs through fees from users. This approach allows the service to be financially sustainable while removing hiring burdens from individual research groups. It delivers significant impact by focusing on short-term projects to help with the "exploitation" phase of AI adoption, after which individual labs can develop deeper research efforts. This allows for rapid response times, as the service can progress from a scoping meeting to answering a research question in just a few weeks. Approximately half of the projects undertaken are ready to start immediately, enabling quick wins for researchers new to AI.

The KU Leuven "AI as a Service" model demonstrates how a targeted intervention can have outsized impact by focusing on removing specific barriers to AI experimentation and enabling researchers to use existing tools to deliver significant value. It illustrates how designing support services around researcher needs, rather than technology push, can accelerate adoption in practice.

HUN-REN AI for Science Initiative[63]

As Hungary's national research network connecting 11 research centres, 8 research institutes, and 116 supported research groups, HUN-REN has developed an AI for Science support model that emphasises capacity building and knowledge sharing.

Embedded in HUN-REN's approach is a three-tiered conceptualisation of AI integration in science: augmented intelligence (enhancing researchers' capabilities), automated intelligence (tasks or processes to be automated), and autonomous intelligence (exploring fundamentally new approaches to scientific discovery). This framework guides the initiative's activities.

HUN-REN's AI Ambassador Network is at the centre of its activities. This network deploys AI specialists across research institutions to provide local support, training, and guidance. Alongside this network, the initiative provides educational resources and access to technological platforms – including HUN-REN's cloud infrastructure – creating a support ecosystem that researchers can access based on their specific needs. The initiative has also launched an AI-First Science programme that explores how AI might fundamentally transform scientific workflows.

In three months, HUN-REN's AI Support Centre supported eight institutional workshops and delivered 14 projects. During the same period, AI Ambassadors provided personal consultation to 126 researchers, engaged 226 researchers in workshops, and involved 511 researchers in locally organised educational programmes. This extensive reach demonstrates the effectiveness of the distributed support model.

The HUN-REN initiative shows how organisational innovation can leverage existing human resources to deliver impact at scale.

Box 2: Case studies of institutional initiatives to support AI in science

The diversity of approaches represented in these case studies suggests that there is no one-size-fits-all solution for institutionalising AI in science. What emerges clearly from all five examples, however, is that successful AI adoption in science requires coherent, multi-faceted funding interventions. Those interventions need to address the technical, human, and cultural dimensions of AI in science, and allow local flexibility in how barriers to adoption are overcome. In response, policymakers engaged in this MLE considered the following recommendations:

[63] https://hun-ren.hu/research_news/ai4science-programme-supported-projects-108067.

- **Develop comprehensive funding portfolios** that balance strategic investments in infrastructure with flexible mechanisms supporting bottom-up experimentation and researcher-led innovation.
- **Balance investments between technical and human infrastructure**, ensuring funding for software engineers, AI specialists, and data managers. This might include targeted institutional support mechanisms like AI service models and ambassador networks that provide flexible access to AI expertise at scale.
- **Design funding instruments that enable interdisciplinary collaboration** by creating evaluation frameworks that reward genuine cross-disciplinary work; supporting both exploitation of known AI methods and exploration of novel approaches; and funding dedicated positions that bridge AI expertise and domain knowledge.

3. Framework Conditions for Responsible AI Integration

3.1. From ethical principles to practice

Over the last 10 years, EU policy initiatives have advanced a European vision for ethical AI, articulated in the EU's Principles for Trustworthy AI. These principles identify seven key requirements that AI systems should demonstrate to be considered trustworthy, setting the contours for acceptable use. They emphasise the importance of human agency and oversight; technical robustness and safety; data privacy and appropriate data governance; transparency; diversity, non-discrimination and fairness; and societal and environmental wellbeing.[64]

Wider adoption of AI in science brings with it new concerns about the ethical implications of AI use in science, and questions about how AI adoption in science can be steered to adhere to these ethical principles. In a recent survey of scientists about their attitudes to AI, most respondents reported both a desire to use AI more in their everyday work and concerns about its responsible use: 67% expressed interest in the use of AI to review large amounts of information, for example, to review literature or analyse data; 54% cited ethical concerns as a barrier to AI adoption.[65]

Ethical issues in relation to the use of AI in science include:[66]

- **Data governance:** AI systems can amplify biases present in data, leading to inaccurate research findings. Re-use of data for purposes not originally specified can undermine research participant consent. Analysis of personal information – including sharing of such information with externally-sourced AI tools – could create privacy and security risks. Each of these issues raises ethical concerns in the context of AI in science.
- **Reproducibility:** Reproducibility is a central pillar of scientific credibility. The complexity of AI models, coupled with issues associated with their accuracy and transparency, has brought new concerns about the impact of AI in the reproducibility of scientific results.[67] The interdependence of data management, AI methods, and research outcomes adds complexity to the challenge of ensuring reproducible science.

[64] https://digital-strategy.ec.europa.eu/en/policies/expert-group-ai.
[65] Nature News (2025), How are researchers using AI? Survey reveals pros and cons for science, https://www.nature.com/articles/d41586-025-00343-5.
[66] Resnik, D.B. and Hosseini, M. (2024), The ethics of using artificial intelligence in scientific research: new guidance needed for a new tool, *AI Ethics*, https://doi.org/10.1007/s43681-024-00493-8.
[67] Ball, P. (2023), Is AI leading to a reproducibility crisis in science? Nature 624, 22-25 (2023) doi: https://doi.org/10.1038/d41586-023-03817-6.

- **Transparency and accountability:** The complexity of some – so-called 'black box' – AI models means that it can be challenging to explain how or why they reach a particular result or conclusion. Generative AI can produce "careless speech"[68] – content that appears authoritative but contains subtle errors that require domain expertise to detect. This acts as a barrier to scientific understanding and increases concerns about accountability for the outputs from AI-enabled studies. Researchers and institutions need clear protocols to ensure humans remain accountable for the results of AI enabled analysis.

While AI is influencing how research is carried out, research integrity norms have not changed. The principles and ways of working set out in the European Code of Conduct for Research Integrity, for example, continue to apply to AI-enabled science. The Code proposes that the following four principles should guide researchers in their work:[69]

- *"Reliability in ensuring the quality of research, reflected in the design, the methodology, the analysis and the use of resources.*
- *Honesty in developing, undertaking, reviewing, reporting and communicating research in a transparent, fair, full and unbiased way.*
- *Respect for colleagues, research participants, society, ecosystems, cultural heritage and the environment.*
- *Accountability for the research from idea to publication, for its management and organisation, for training, supervision and mentoring, and for its wider societal impacts."*[70]

These principles imply that, in AI-enabled science:

- Researchers should disclose their use of AI, as part of their research to honestly and transparently describe their work.
- Human authors are fully responsible and accountable for their work, and should take care to avoid bias, inaccuracies, and hallucinations arising from AI.
- Caution is needed about the use of non-institutional AI tools for analysis of sensitive data, due to the privacy or security implications, if third parties access that data.

These principles are reflected in the 2024's European Commission-published 'Living Guidelines on the Responsible Use of Generative AI in Research'.[71] These guidelines make recommendations to researchers, research organisations, and funding bodies, including that:

- Researchers should maintain responsibility for AI-enabled research outputs. They should be transparent about where AI has been used in research, take action to protect sensitive research data when using AI, and consider the ethical implications of their use of AI technologies.
- Research organisations should support the responsible use of AI in research through the provision of guidelines, training, and sharing of good practices. They should provide infrastructure for responsible AI use and take steps to monitor how AI is being adopted.
- Funding bodies should design funding instruments to enable responsible AI in research and encourage transparency in its use.

[68] Wachter, S., Mittelstadt, B., and Russell, C. (2024), Do large language models have a legal duty to tell the truth? (January 31, 2024). Royal Society Open Science,

SSRN: https://ssrn.com/abstract=4771884 or http://dx.doi.org/10.2139/ssrn.4771884.

[69] Code of Conduct available at : https://allea.org/code-of-conduct/.

[70] https://allea.org/european-code-of-conduct-2017/.

[71] Guidelines on the responsible use of generative AI in research available at: https://research-and-innovation.ec.europa.eu/news/all-research-and-innovation-news/guidelines-responsible-use-generative-ai-research-developed-european-research-area-forum-2024-03-20_en.

While helpful in starting a conversation about how scientists can uphold standards for research integrity when using AI, there is a gap between these high-level ethical principles and the practice of AI in science. Researchers often struggle to translate the aspirations set out in ethical frameworks to practical implementation and need interventions that make these principles tangible. One example of such an approach comes from the Flanders AI Research programme, which set out to create spaces for critical reflection on the implications of AI for science as part of its work to promote the adoption of AI in science by:

- Raising awareness of the importance of AI ethics in science, with an internal responsible AI resource page for all researchers in the programme;
- Creating a responsible AI self-assessment tool to support self-reflection;
- Convening a cross-cutting responsible AI research programme that fed into its wider research portfolio, for example bringing together 200 researchers in a futures workshop to examine emerging issues.

These experiences show that formal policies alone will not be sufficient to drive responsible AI adoption in science, also because the pace of technological change is extremely fast. Embedding established norms from research integrity and ethics in AI in science requires researchers to be prepared to examine the implications of using AI and institutions to provide the social and technical resources to enable responsible AI implementation. This might include:

- Training for researchers and research managers in best practices for AI in science;
- Secure data environments that are connected to HPC facilities;
- Specialist data managers and software engineers to manage data resources and develop open software tools;
- Clear protocols for AI use and documentation standards;
- Research cultures that encourage open sharing of best practices.

3.2. Open science as an enabler of AI in science

Open science frameworks can also support responsible innovation in AI in science, by increasing the transparency and accessibility of AI in science research. Key elements of these frameworks include:

- **Open access publications:** Making research freely available accelerates knowledge translation across disciplines and helps promote the adoption of successful methods.
- **Open data initiatives:** Data sharing provides the foundation for training advanced AI models. For example, the Protein Data Bank (PDB),[72] established in 1971, is an open repository for data about biological structures.[73] The PDB made protein structure data widely available for research through a Creative Commons open-source license.[74] The result is a vital public data resource that enabled breakthroughs like AlphaFold.
- **Open-source software development:** Open-source software development supports the verification of AI methods and reproducibility of scientific results. For example, CASTEP, a modelling code that calculates the properties of materials from first principles, was developed at the University of Cambridge and is distributed as open-source software. It has been used

[72] https://www.rcsb.org/.

[73] Berman, H.M., Westbrook, J., Feng, Z. et al. (2000), The Protein Data Bank. Nucleic Acids Research, 28(1), 235-242, https://doi.org/10.1093/nar/28.1.235.

[74] https://www.rcsb.org/news/feature/611e8d97ef055f03d1f222c6.

in a range of research projects, supporting advances in computational chemistry and materials discovery.[75]

These aspects of open science are supported in multiple European policy frameworks, including Open Science policies under Horizon Europe; the European Open Science Cloud (EOSC); and Open Research Europe.[76] Similarly to research integrity, researchers need practical interventions and resources to translate these frameworks to practice.

3.3. Anticipatory governance for AI in science

AI capabilities, patterns of AI adoption in science, and understandings of its implications for scientific practice are in flux. The result is that policymakers are operating in conditions of high uncertainty and rapid change, where traditional approaches to technology governance struggle to keep pace with technological development. In this context, anticipatory governance can play a role in supporting policy to respond to new opportunities or challenges in AI in science.

Anticipatory governance is a framework for policymaking that emphasises foresight, experimentation, and continuous learning in a dynamic policy framework.[77] Its aim is to help decision-makers anticipate and prepare for new opportunities or threats.[78] The result should be greater agility in policymaking.

To adopt an anticipatory approach, policymakers need the ability to understand the innovation landscape and how it is changing, to think systematically about how to mobilise the innovation ecosystem in response, and the capacity to support experimentation in a test-and-learn approach to policymaking.[79] This relies on:

- Intelligence gathering to identify emerging issues that may require policy attention and to track the impact of policy interventions.
- Experimentation and evaluation during policy implementation, tracking which interventions are working and establishing feedback loops that identify where different responses are needed.
- Horizon scanning and foresight activities that convene diverse stakeholders to discuss emerging developments, looking for signals of potential risks or opportunities.

One example of how to create such an intelligence function comes from Flanders' AI Research Programme. This programme has convened futures workshops with researchers from across its networks, with the aim of identifying signals or trends in research that might have policy implications. Its AI Steering Committee brings together representatives from science, ethics, law, industry, and academia, who meet quarterly to consider emerging issues. These spaces for reflection with diverse stakeholders can help institutions anticipate and adapt to emerging developments.

Drawing from the OECD's framework for anticipatory governance[80] and examples of research networks and policy interventions explored during the MLE, Box 3 sketches an anticipatory governance function for AI in science.

[75] https://www.castep.org/.

[76] https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/open-science_en.

[77] https://www.oecd.org/en/topics/anticipatory-governance.html.

[78] https://www.weforum.org/stories/2025/03/anticipatory-governance-for-a-risky-world/.

[79] UNDP (2020), Anticipatory governance – a primer, https://www.undp.org/vietnam/blog/anticipatory-governance-primer.

[80] https://www.oecd.org/en/topics/anticipatory-governance.html.

Anticipatory governance aims to create learning policy systems. In AI for science, the following functions could contribute to such a system:

Establishing an intelligence network:

- Expert panels and forums: Convene diverse networks of AI researchers, domain scientists, ethicists, policy experts, and industry representatives – including different career stages – to contribute fresh perspectives on emerging trends and challenges.
- International comparisons: Build connections with equivalent bodies in other countries to share insights, approaches and experiences.
- Public dialogues: Engage with diverse publics through structured consultations and workshops to understand societal concerns and values.

Integrating horizon-scanning in policymaking:

- Foresight exercises: Develop structured scenario planning and futures workshops to understand how AI might transform different scientific fields.
- Monitoring and signal detection: Track developments in AI research, scientific applications, and governance frameworks, for example by systematically reviewing publications for new methods or applications.
- Connect with policymaking: Bring insights from foresight exercises into policymaking, for example through sandboxes.

Creating spaces for policy experimentation:

- Sandboxes: Establish spaces to experiment with governance approaches for novel AI applications in science or to test the implications of new AI in science applications.
- Prototyping: Deploy agile policymaking methods to test interventions and scale successful approaches.
- Feedback loops: Create structured mechanisms to evaluate and improve policy interventions.

Box 3: Designing an anticipatory governance function for AI in science[81]

Discussions at MLE workshops highlighted the following recommendations for policymakers to build capacity in the governance of AI in science:

- **Target the next phase of policy interventions** to bridge the implementation gap between governance frameworks and practice (for example, In 'Findable Accessible Interoperable Reusable' (FAIR) data or research culture).
- **Create practical implementation tools** like self-assessment guides, documentation standards, and training programmes that translate ethical principles to scientific workflows.
- **Establish anticipatory governance mechanisms** including foresight activities and stakeholder engagement processes that look for signals of emerging challenges or opportunities and provide feedback loops on the effectiveness of policy interventions.

[81] Insights in this box draw from discussions at the MLE workshop, practices in anticipatory governance, and activities convened by expert participants.

4. Taking an Ecosystem Approach

Many of the framework conditions required to support progress in AI in science are shared across different areas of research, for example: research funding, responsible innovation, open science, and research culture. AI in science also shares a collection of infrastructure needs with many areas of AI research, including the need for large-scale HPC, access to data, and availability of software tools.

These policy agendas are connected, as illustrated by the following example. Policymakers considering infrastructure investments encounter a web of interactions between different policies and research activities. HPC facilities, data resources, and software tools provide technical building blocks for AI in science research. Access to such infrastructure is an attractor for AI talent and can provide a convening point for interdisciplinary research programmes. However, compute alone is not sufficient for AI adoption in science. Adoption requires talent that can make effective use of HPC facilities. This in turn requires investment in education, talent development, and specialist roles that provide the human capital needed to leverage these technical resources. This might include knowledge brokers, training programmes, and ambassador networks to spread expertise in AI for science, as the ability to readily access specialised engineering support lowers the barrier to entry for researchers to engage with AI in science. As a result, strategic investments in both technical and human infrastructure are necessary to create a foundation for the adoption of AI in science.

Table 2[82] summarises the range of policy elements that have been described during the MLE on national policies for AI in science. These highlight:

- There is a set of 'policy essentials' for AI in science, which include:
  - Data governance mechanisms that establish clear protocols for data sharing and management.
  - Skills-building and talent-development policies that educate researchers in AI and give attractive career pathways for AI in science specialists.
  - Implementation support, for example through software engineering teams, that help researchers deploy AI in their research.
  - Funding frameworks that encourage cross-disciplinary collaboration.
  - HPC infrastructures designed for AI in science applications.
  - Research governance interventions that embed responsible innovation in AI in science.
- Enablers of success include:
  - Clear policy objectives, rooted in a vision for AI in science that has public support and that maintains scientific values.
  - Effective stakeholder engagement across science, industry, and policy.
  - Research cultures that celebrate interdisciplinary work.
  - Communities of practice that support knowledge-sharing and scaling of successful approaches across institutions.
  - Opportunities to experiment and learn from what works in practice.
  - Policy frameworks that evolve with technological developments.

[82] This table was generated with help from AI, based on notes taken during MLE workshops.

Core Dimensions	Subdimensions	Policy Elements
1. Framework conditions	**Funding**	• Funding portfolios that combine both strategic investments in infrastructure and bottom-up experimentation support that helps researchers deploy AI in their science. • Funding progression pathways from seed projects to larger initiatives. • Public-private partnership models with appropriate governance frameworks. • Flexible funding instruments that accommodate rapid evolution of AI technologies.
	Governance	• Ethical frameworks and research governance structures. • Practical ethics or governance implementation tools like self-assessment guides. • Anticipatory governance mechanisms to identify and respond to emerging risks. • Domain-specific guidance for responsible AI use in different scientific fields that reflects the needs or issues faced by researchers in those domains.
	Research culture	• Research assessment frameworks that recognise interdisciplinary contributions. • Incentives for knowledge sharing and collaboration across disciplines. • Spaces for experimentation with novel AI approaches. • Recognition schemes for cross-disciplinary publications and innovations.
2. Infrastructure	**Computational power**	• Computing resources with appropriate interfaces for scientific users. • Energy-efficient computing practices and infrastructure.
	Data	• FAIR data principles used as standard in research. • Clear data governance frameworks that enable research while protecting privacy. • Resources to improve data quality.

Core Dimensions	Subdimensions	Policy Elements
	Software	• Funding for development and maintenance of open-source scientific AI tools (for example, software toolkits, code libraries, open source models). • Documentation standards for AI use in scientific research. • Reproducibility frameworks for AI-enabled scientific discoveries (for example, code versioning and documentation, data provenance tracking, and methods for output validation).
3. Human capital and adoption	**Communities**	• Technical support roles like research software engineers and data managers. • Knowledge broker positions to connect AI expertise with domain knowledge. • AI service models like AI-as-a-Service and ambassador networks. • Communities of practice, networking events, and collaborative platforms.
	Talent	• Interdisciplinary training programmes that bridge AI and domain sciences. • Dedicated career pathways for AI-science professionals. • Specialised roles for AI support functions in research organisations. • AI literacy integrated into scientific education at all levels. • Investment to attract and retain AI talent and to nurture talent in AI in science.
	Adoption approaches	• Support for varied AI adoption modes (augmented, automated, autonomous intelligence). • Create implementation guides and pathways for different research contexts. • Translation mechanisms to move AI innovations between fields. • Community-led spaces for innovation and experimentation. • Networks to improve interdisciplinary understanding and incubate collaborations.

Table 2: Policy elements from the MLE process

A consequence of this diversity of interventions is that AI for science sits at the intersection of multiple policy domains, including: AI policies, R&I policies, education policies, and industrial policies. Who is responsible for these different policy agendas varies across countries, but they are typically managed by different ministries or agencies. For example, AI policy led by digital affairs ministries, research policy by education or science ministries, growth-oriented policies by economic ministries. This may create challenges around cross-government coordination and policy coherence. Effective coordination is needed to prevent gaps or misalignment in support structures for AI in science.

Based on this understanding of how different elements interact and reinforce each other, participants in MLE workshops and literature reviews identified several recommendations that emerge for policymakers seeking to build effective AI for science ecosystems:

- **Establish coordination mechanisms** across government departments responsible for AI, research, infrastructure, and education that together provide a coherent framework for AI in science.
- **Support communities of practice** that foster knowledge-sharing across disciplinary boundaries, providing frameworks that allow for local adaptation and innovation.
- **Design policy feedback loops** through monitoring, evaluation, and regular stakeholder engagement.

5. Conclusion

The use of AI for scientific discovery is both an opportunity to drive a new wave of innovation and a complex policy challenge. Creating an enabling environment for AI in science requires coordinated action across funding frameworks, governance mechanisms, infrastructure development, and talent strategies. These policy domains are interconnected, and policy development is taking place in an environment where technological capabilities and research practices are shifting.

Successful AI in science initiatives share common characteristics that can inform future policy development:

- **Pairing strategic investments with bottom-up experimentation:** Effective policy portfolios balance large-scale, coordinated investments with flexible mechanisms that enable researchers to respond to opportunities in their domains and provide support structures that are adapted to local contexts.
- **Building human and technical infrastructures:** While compute resources and data platforms provide the technical foundation for AI in science, investments in interdisciplinary researchers, software engineers, knowledge brokers, and institutional support mechanisms are equally critical for widespread adoption.
- **Bridging policy to practice:** Translating high-level ethical principles into research workflows requires practical tools, training, and communities of practice that help researchers navigate the ethical and technical challenges of AI in science.
- **Anticipating change:** Policy mechanisms that incorporate foresight activities, stakeholder engagement, and continuous learning help align governance with evolving technologies and practices.

Policymakers engaged in this MLE considered success in terms of Europe's ability to "*do more science, better*", with the aim of leveraging AI to accelerate scientific progress and translate this progress to beneficial economic and social impact. Such translation might be through scientific outputs and forms of knowledge exchange across science and industry, or through the use of new knowledge to catalyse spin-out and start-up activities in Europe.

Box 4 summarises policy recommendations considered across this report. Drawing from insights shared during the MLE, Figure 1 proposes a framework for AI in science policy, illustrating the range of policy levers available to policymakers seeking to promote the adoption of AI in science. The challenge for the next wave of policy development in this area will be to bridge across policy and practice, delivering innovative interventions that enhance AI adoption in science. An over-arching recommendation is that policymakers consider how to bring together the diverse policy levers and frameworks considered in this report to deliver a national ambition for AI in science.

Below is a summary of policy areas for action considered by participants in the MLE workshop on framework conditions and funding for AI in science:

- Develop comprehensive funding portfolios that balance strategic investments in infrastructure with flexible mechanisms supporting bottom-up experimentation and researcher-led innovation.
- Balance investments between technical and human infrastructure, ensuring funding for software engineers, AI specialists, and data managers. This might include targeted

institutional support mechanisms like AI service models and ambassador networks that provide flexible access to AI expertise at scale.

- Design funding instruments that enable interdisciplinary collaboration by creating evaluation frameworks that reward genuine cross-disciplinary work; supporting both exploitation of known AI methods and exploration of novel approaches; and funding dedicated positions that bridge AI expertise and domain knowledge.
- Target the next phase of policy interventions to bridge the implementation gap between governance frameworks and practice (for example, In 'Findable Accessible Interoperable Reusable' (FAIR) data or research culture).
- Create practical implementation tools like self-assessment guides, documentation standards, and training programmes that translate ethical principles to scientific workflows.
- Establish anticipatory governance mechanisms including foresight activities and stakeholder engagement processes that look for signals of emerging challenges or opportunities and provide feedback loops on the effectiveness of policy interventions.
- Establish coordination mechanisms across government departments responsible for AI, research, infrastructure, and education that together provide a coherent framework for AI in science.
- Support communities of practice that foster knowledge-sharing across disciplinary boundaries, providing frameworks that allow for local adaptation and innovation.
- Design policy feedback loops through monitoring, evaluation, and regular stakeholder engagement.

Box 4: Summary of recommendations

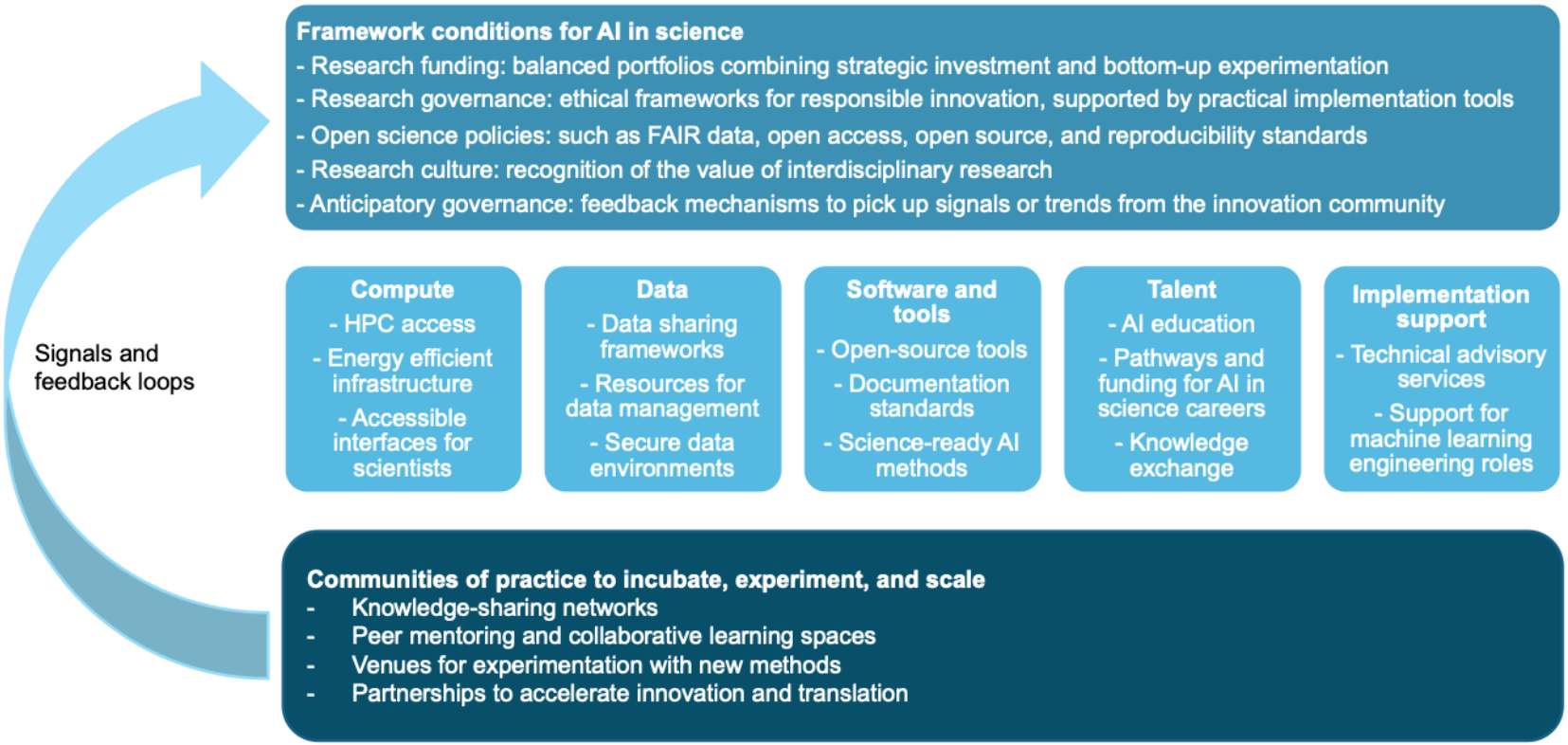


Figure 1: A framework for AI in science policy
Source: Author's own creation

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The use of AI for scientific discovery is both an opportunity to drive a new wave of innovation and a complex policy challenge. Creating an enabling environment for AI in science requires coordinated action across funding frameworks, governance mechanisms, infrastructure development, and talent strategies. These policy domains are interconnected, and policy development is taking place in an environment where technological capabilities and research practices are shifting. Drawing from discussions at the MLE workshop in Belgium on 20-21 March 2025, a survey of national policymakers, and desk-based research, this report explores successful policy and funding models for AI in science from across Europe and proposes recommendations to create an enabling environment for AI in science innovation.

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